

TFB2063:Data Science - May 2026

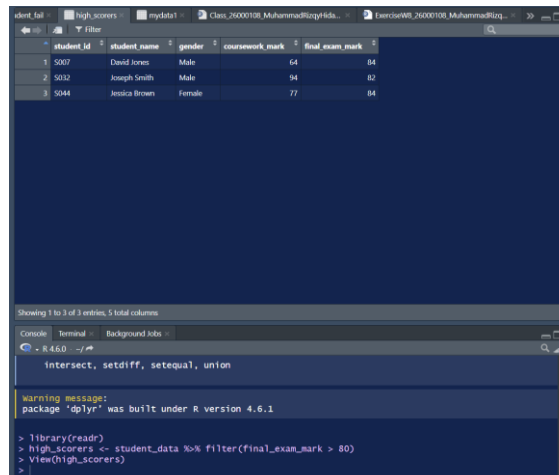
Week 8 Exercise

1. Show the details of students whose final_exam_mark is greater than 80%.

Code:

```
high_scorers <- student_data %>% filter(final_exam_mark > 80)  
View(high_scorers)
```

Result:



The image shows the RStudio interface. At the top, the 'Environment' pane displays a data frame named 'student_data' with 5 columns: student_id, student_name, gender, coursework_mark, and final_exam_mark. The 'Data' pane shows the first three rows of this data frame.

student_id	student_name	gender	coursework_mark	final_exam_mark	
1	5007	David Jones	Male	64	84
2	5032	Joseph Smith	Male	94	82
3	5044	Jessica Brown	Female	77	84

Below the data frame, the 'Console' pane shows the following R code and output:

```
> library(readr)
> high_scorers <- student_data %>% filter(final_exam_mark > 80)
> View(high_scorers)
>
```

The output of the code is a warning message:

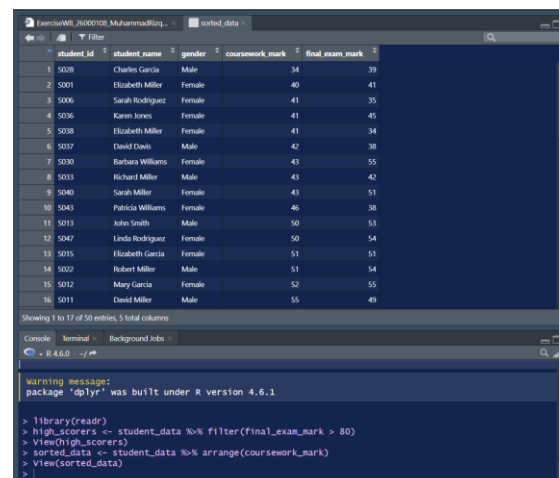
```
Warning message:
package 'dplyr' was built under R version 4.6.1
```

2. Arrange the dataset in ascending order based on course_work_mark.

Code:

```
sorted_data <- student_data %>% arrange(coursework_mark)  
View(sorted_data)
```

Result:



The screenshot shows the RStudio interface. The top pane displays a data frame with 16 rows and 5 columns: student_id, student_name, gender, coursework_mark, and final_exam_mark. The data is sorted by coursework_mark in ascending order. The bottom pane shows the R console with a warning message indicating that the dplyr package was built under R version 4.6.1.

student_id	student_name	gender	coursework_mark	final_exam_mark	
1	5038	Charles Garcia	Male	34	39
2	5001	Elizabeth Miller	Female	40	41
3	5006	Sarah Rodriguez	Female	41	35
4	5036	Karen Jones	Female	41	45
5	5038	Elizabeth Miller	Female	41	34
6	5037	David Davis	Male	42	38
7	5030	Barbara Williams	Female	43	55
8	5033	Richard Miller	Male	43	42
9	5040	Sarah Miller	Female	43	51
10	5043	Patricia Williams	Female	46	38
11	5013	John Smith	Male	50	53
12	5047	Linda Rodriguez	Female	50	54
13	5015	Elizabeth Garcia	Female	51	51
14	5022	Robert Miller	Male	51	54
15	5012	Mary Garcia	Female	52	55
16	5011	David Miller	Male	55	49

Showing 1 to 17 of 50 entries, 5 total columns

```
> library(readr)
> high_scorers <- student_data %>% filter(final_exam_mark > 80)
> View(high_scorers)
> sorted_data <- student_data %>% arrange(coursework_mark)
> View(sorted_data)
>
```

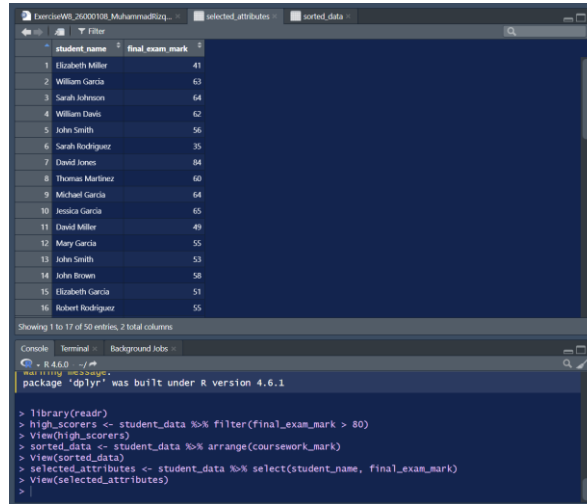
Warning message:
package 'dplyr' was built under R version 4.6.1

3. Display only the student_name and final_exam_mark attributes from the dataset.

Code:

```
selected_attributes <- student_data %>% select(student_name,  
final_exam_mark)  
View(selected_attributes)
```

Result:



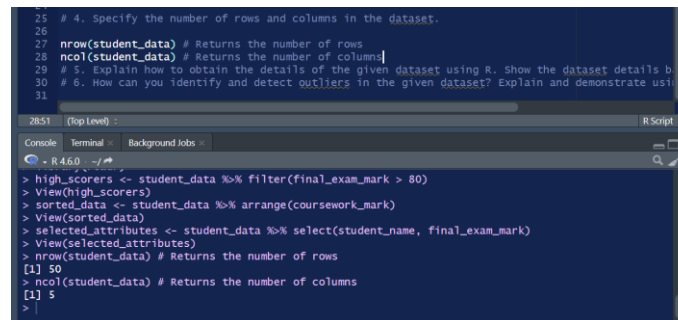
	student_name	final_exam_mark
1	Elizabeth Miller	41
2	William Garcia	63
3	Sarah Johnson	64
4	William Davis	62
5	John Smith	56
6	Sarah Rodriguez	35
7	David Jones	84
8	Thomas Martinez	60
9	Michael Garcia	64
10	Jessica Garcia	65
11	David Miller	49
12	Mary Garcia	55
13	John Smith	53
14	John Brown	58
15	Elizabeth Garcia	51
16	Robert Rodriguez	55

4. Specify the number of rows and columns in the dataset.

Code:

```
nrow(student_data) # Returns the number of rows  
ncol(student_data) # Returns the number of columns
```

Result:



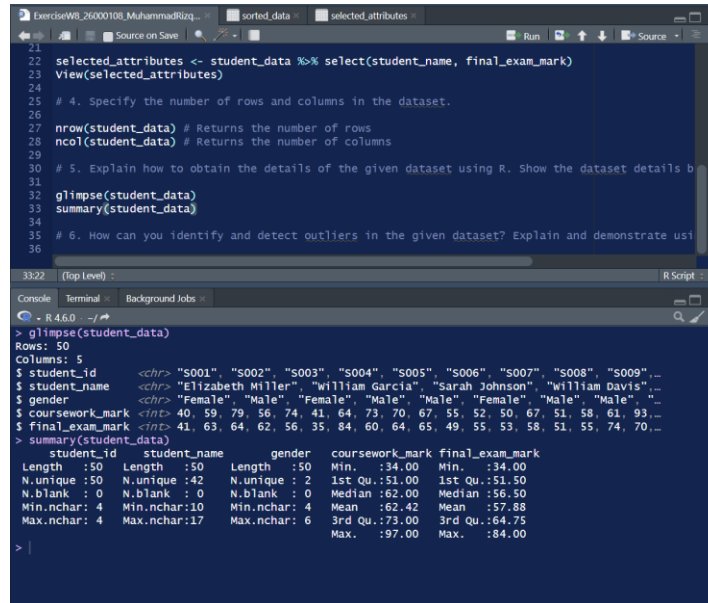
```
25 # 4. Specify the number of rows and columns in the dataset.  
26  
27 nrow(student_data) # Returns the number of rows  
28 ncol(student_data) # Returns the number of columns  
29  
30 # 5. Explain how to obtain the details of the given dataset using R, show the dataset details b  
31 # 6. How can you identify and detect outliers in the given dataset? Explain and demonstrate usi  
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```

5. Explain how to obtain the details of the given dataset using R. Show the dataset details based on the R query you execute.

Code:

```
glimpse(student_data)
summary(student_data)
```

Result:



```
selected_attributes <- student_data %>% select(student_name, final_exam_mark)
View(selected_attributes)

# 4. Specify the number of rows and columns in the dataset.
nrow(student_data) # Returns the number of rows
ncol(student_data) # Returns the number of columns

# 5. Explain how to obtain the details of the given dataset using R. Show the dataset details b
glimpse(student_data)
summary(student_data)

# 6. How can you identify and detect outliers in the given dataset? Explain and demonstrate usi
```

```
> glimpse(student_data)
#> # A tibble: 50 x 5
#>   student_id student_name gender coursework_mark final_exam_mark
#>   <chr>      <chr>      <chr>      <dbl>          <dbl>
#> 1 "S001"     "Elizabeth Miller" "Female"    40.00          41.00
#> 2 "S002"     "William Garcia"   "Male"     59.00          63.00
#> 3 "S003"     "Sarah Johnson"    "Female"   79.00          64.00
#> 4 "S004"     "William Davis"    "Male"     56.00          56.00
#> 5 "S005"     "Sarah Johnson"    "Female"   74.00          62.00
#> 6 "S006"     "William Davis"    "Male"     41.00          35.00
#> 7 "S007"     "Elizabeth Miller" "Female"   64.00          84.00
#> 8 "S008"     "Sarah Johnson"    "Female"   67.00          60.00
#> 9 "S009"     "William Davis"    "Male"     55.00          49.00
#> 10 "S010"     "Elizabeth Miller" "Female"   52.00          55.00
#> 11 "S011"     "William Davis"    "Male"     50.00          53.00
#> 12 "S012"     "Sarah Johnson"    "Female"   67.00          58.00
#> 13 "S013"     "Elizabeth Miller" "Female"   51.00          55.00
#> 14 "S014"     "William Davis"    "Male"     58.00          74.00
#> 15 "S015"     "Sarah Johnson"    "Female"   93.00          70.00
#> 16 "S016"     "Elizabeth Miller" "Female"   40.00          41.00
#> 17 "S017"     "William Garcia"   "Male"     59.00          63.00
#> 18 "S018"     "Sarah Johnson"    "Female"   79.00          64.00
#> 19 "S019"     "William Davis"    "Male"     56.00          56.00
#> 20 "S020"     "Sarah Johnson"    "Female"   74.00          62.00
#> 21 "S021"     "William Davis"    "Male"     41.00          35.00
#> 22 "S022"     "Sarah Johnson"    "Female"   64.00          84.00
#> 23 "S023"     "Elizabeth Miller" "Female"   67.00          60.00
#> 24 "S024"     "William Davis"    "Male"     55.00          49.00
#> 25 "S025"     "Elizabeth Miller" "Female"   52.00          55.00
#> 26 "S026"     "William Davis"    "Male"     50.00          53.00
#> 27 "S027"     "Sarah Johnson"    "Female"   67.00          58.00
#> 28 "S028"     "Elizabeth Miller" "Female"   51.00          55.00
#> 29 "S029"     "William Davis"    "Male"     58.00          74.00
#> 30 "S030"     "Sarah Johnson"    "Female"   93.00          70.00
#> 31 "S031"     "Elizabeth Miller" "Female"   40.00          41.00
#> 32 "S032"     "William Garcia"   "Male"     59.00          63.00
#> 33 "S033"     "Sarah Johnson"    "Female"   79.00          64.00
#> 34 "S034"     "William Davis"    "Male"     56.00          56.00
#> 35 "S035"     "Sarah Johnson"    "Female"   74.00          62.00
#> 36 "S036"     "William Davis"    "Male"     41.00          35.00
#> 37 "S037"     "Sarah Johnson"    "Female"   64.00          84.00
#> 38 "S038"     "Elizabeth Miller" "Female"   67.00          60.00
#> 39 "S039"     "William Davis"    "Male"     55.00          49.00
#> 40 "S040"     "Elizabeth Miller" "Female"   52.00          55.00
#> 41 "S041"     "William Davis"    "Male"     50.00          53.00
#> 42 "S042"     "Sarah Johnson"    "Female"   67.00          58.00
#> 43 "S043"     "Elizabeth Miller" "Female"   51.00          55.00
#> 44 "S044"     "William Davis"    "Male"     58.00          74.00
#> 45 "S045"     "Sarah Johnson"    "Female"   93.00          70.00
#> 46 "S046"     "Elizabeth Miller" "Female"   40.00          41.00
#> 47 "S047"     "William Garcia"   "Male"     59.00          63.00
#> 48 "S048"     "Sarah Johnson"    "Female"   79.00          64.00
#> 49 "S049"     "William Davis"    "Male"     56.00          56.00
#> 50 "S050"     "Sarah Johnson"    "Female"   74.00          62.00
```

To obtain the structural and statistical details of a dataset in R, we use two complementary functions: `glimpse()` and `summary()`.

- The `glimpse()` function is provided by the `dplyr` library to see the columns of the dataset and display some portion of the data with respect to each attribute/variable. It is like a transposed output of the standard print function, making it incredibly easy to view every column name and its underlying data type (like numeric or character).
- The `summary()` function is a base R function used to obtain descriptive statistics for the dataset. For numerical attributes, it calculates and displays the minimum, 1st quartile, median, mean, 3rd quartile, and maximum values. This helps quickly identify the distribution of the data and spot potential outliers.

6. How can you identify and detect outliers in the given dataset? Explain and demonstrate using R.

Outlier detection within the `final_exam_mark` variable of the `student_data` dataset is systematically identified utilizing three steps of standard outlier handling: 1. Calculate the Quartiles and IQR; 2. Establish the Outlier Boundaries; 3. Display the calculated data

To rigorously classify these anomalies, the Interquartile Range (IQR) methodology is applied. This technique defines outliers based on the statistical dispersion of the middle 50% of the dataset.

1. Calculate the Quartiles and IQR

Determine the 25th percentile (first quartile, Q1) and 75th percentile (third quartile, Q3) using the `quantile()` function. Subsequently, the IQR is derived to quantify the spread:

Code:

```
first_q <- quantile(student_data$final_exam_mark, 0.25)
third_q <- quantile(student_data$final_exam_mark, 0.75)
iqr_val <- IQR(student_data$final_exam_mark)
```

2. Establish the Outlier Boundaries

Mathematical boundaries, termed the lower extreme and upper extreme, are then established. By standard statistical convention, any data point falling below **Q1 - (1.5 * IQR)** or exceeding **Q3 + (1.5 * IQR)** is formally classified as an outlier. These thresholds are computed in R as follows:

Code:

```
le <- first_q - 1.5 * iqr_val
ue <- third_q + 1.5 * iqr_val
```

3. Display the calculated data

By querying the dataset against these defined `le` and `ue` parameters, aberrant observations are systematically isolated. This identification allows the system to apply appropriate data preprocessing interventions, such as observation exclusion or mean imputation, prior to further analysis.

```
> # Calculate the Quartiles and IQR
> first_q <- quantile(student_data$final_exam_mark, 0.25)
> third_q <- quantile(student_data$final_exam_mark, 0.75)
> iqr_val <- IQR(student_data$final_exam_mark)
> # Calculate boundaries
> le <- first_q - 1.5 * iqr_val
> ue <- third_q + 1.5 * iqr_val
> # View the boundaries
> le
25%
31.625
> ue
75%
84.625
>
```